

FARM MANAGEMENT SYSTEM

2. REQUIREMENT ANALYSIS

2.1 Project Overview

Requirement analysis identifies the functions and features required for the Farm Management System. The system is designed using Salesforce Lightning to manage farmer, farm, crop and buyer information. It includes data validation, object relationships, user access control, reports and dashboards for effective farm management.

2.2 Functional Requirements

ID	Requirement	Description
FR-1	User Authentication	Provide secure access to authorized users.
FR-2	Farm, Crop, Farmer and Buyer Management	Maintain farm details such as location, size, type and owner information.
FR-3	Data Entry and Validation	Record and manage crop details, production and harvesting information.
FR-4	Relationship Mapping	Establish relationships between farmers, farms, crops and buyers.
FR-5	Report Generation	Generate reports related to farmers, farms, crops, buyers and sales.
FR-6	Dashboard Visualization	Display important farm information through charts and dashboards.
FR-7	User Role Management	Control access to records according to user roles and permissions.

2.3 Non-Functional Requirements

Category	Requirement
Usability	Provide a simple Salesforce Lightning user interface.
Security	Use profiles, roles and sharing controls for access management.
Reliability	Use cloud storage and platform capabilities to reduce the risk of data loss.
Performance	Target report/query generation and loading within approximately 3 seconds.
Availability	Provide cloud-based access for authorized users.
Scalability	Support growth in users, farms, records and geographic coverage.

2.4 Data Flow Diagram (Logical Flow)

User → Salesforce Lightning UI → Farmer/Farm/Crop/Buyer Objects → Data Validation → Object Relationships → Reports → Dashboards

2.5 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend/Application Platform	Salesforce Platform
Database	Salesforce Cloud
Automation/Validation	Salesforce Validation Rules and automation capabilities
Analytics	Salesforce Reports and Dashboards

2.6 Main Data Entities

- **Farmer:** Stores farmer name, contact details, location and other farmer-related information.

- **Farm:** Stores farm location, farm type, size, soil quality, irrigation details and farmer relationship.
- **Crop:** Stores crop name, planting date, harvesting date, yield and related farm information.
- **Buyer:** Stores buyer name, contact details, email, address and buyer type.
- **Sales:** Stores crop sales and buyer-related transaction information.

2.7 Requirement Summary

The requirements establish a centralized Salesforce-based Farm Management System for managing farmers, farms, crops, buyers and sales information. The system provides secure data management, validation, relationships, reports and dashboards to improve the efficiency of farm-related activities. These requirements provide the foundation for the system design and implementation phases.